

Ultimate Nutrition

Glucosamine Sulfate

Millions of Americans, and even greater numbers worldwide, suffer from the discomfort and pain of arthritis. It is the persistent pain, which ultimately results in loss of joint mobility and function that makes arthritis such a debilitating disease. Arthritis literally means a "fire in the joints." This "fire" is because of inflammation in the joint. Although there are numerous forms of arthritis, two most common ones are osteoarthritis and rheumatoid arthritis. Osteoarthritis is most prevalent in people over the age of 45 years, and is a result of age-related wear and tear of the joints. Rheumatoid arthritis is largely hereditary and affects people much earlier in life. The disease process in both rheumatoid and osteo-arthritis is the same.

Presently, there is no cure for arthritis, even though scientists and clinicians have gained considerable insights into the basic processes that contribute to the disease. Fortunately, natural remedies are available that can relieve pain and allow joint mobility and, as a result, mobility and personal autonomy. Glucosamine sulfate is the best-studied natural remedy.

In Europe, a number of clinical studies have been done with glucosamine, which is effective in osteoarthritis. Glucosamine is generally described as "chondro-protective" agent, since it protects the cartilage (chondro) against the degeneration of the joint as a result of arthritis. Cartilage is a specialized type of connective tissue that provides support for the joints, and helps in the joint movement. As such, it practically functions as a shock absorber for the joint.

To understand how glucosamine helps in arthritis, let us see how healthy joints work. Healthy joint cartilage requires three things:

- 1) Proteoglycans to attract and hold water;
- 2) Water for lubrication, nourishment and transport; and
- 3) Collagen to keep proteoglycans in place to support cartilage.

Proteoglycans are specialized molecules that have a protein at their core with sulfate as side chains. Proteoglycans are essential for healthy cartilage because they attract and hold water, which lubricates and nourishes the cartilage. Therefore, proteoglycans are the "cement" that holds cartilage together. Proteoglycans break down with age, which contributes to osteoarthritis.

Glucosamine is a building block of proteoglycans. Glucosamine serves two functions: It is both a building block for cartilage and stimulates the production of the so-called chondrocytes. Chondrocytes are a special type of cells that are miniature factories for the production of collagen and proteoglycans. Both collagen and proteoglycans prevent the breakdown of cartilage. In addition, chondrocytes have another important purpose: They remove worn-out cartilage in order for fresh cartilage to be produced.

Glucosamine plays an important role in determining how much proteoglycan is produced by the chondrocytes. If sufficient amount of glucosamine is present, water-carrying proteoglycans will be produced and not only the joint will be properly lubricated but it will also receive proper nutrients to keep it healthy. If lesser amounts of glucosamine are present, clearly, the joint will suffer.

Numerous published studies demonstrate that glucosamine has beneficial effects on joint cartilage. In clinical trials, sufferers of osteoarthritis taking glucosamine as a nutrient reported reduced pain and improved joint function (please see selected references below). What is important is that glucosamine is a safe nutrient that is well tolerated and can be taken continuously for longer-term relief from the crippling pain of those swollen, aching and hurtful joints.

SELECTED REFERENCES

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Rovati et al., "A Large, Randomized, Placebo-Controlled, Double-Blind Study of Glucosamine Sulfate vs. Piroxicam and vs. Their Association on the Kinetics of Symptomatic Effect on Osteoarthritis," *Osteoarthritis Cartilage*: 2 [Suppl. 1], 56-62, 1994
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Vajaradal, Y., "Double-Blind Clinical Evaluation of Intra-Articular Glucosamine in Outpatients with Gonarthrosis," *Clin. Ther.*: 3, 336-343, 1981.
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FREQUENTLY ASKED QUESTIONS

1. How much glucosamine should I take?

The amounts of glucosamine used in clinical trials have routinely been three dosages per day of 500 mg. That is, 1500 mg of glucosamine each day is the standard amount, which has proven to be effective in numerous clinical trials.

It should be noted, however, that individuals who are taking diuretics may need to increase the dosage to compensate for its reduced effectiveness.

2. Are there any side effects?

In clinical trials, glucosamine has been found to have no side effects and it is well tolerated. In animal experiments, the safety of glucosamine has been demonstrated.

3. Can I take it with other medications?

As noted above, people on diuretics should increase the dosage. In case of prescribed

medications, your primary care physician may be the best source of advice.